Factorial Experiments

Main effects, simple effects and interactions; 2ⁿ and 3ⁿ series, Yates' method and confounding.

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1 The idea

A **factorial experiment** studies two or more factors **simultaneously**, using all combinations of their levels.

Treatment combinations

$$N = \prod s_i$$

Series

$$2^n,\ 3^n$$

$2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $3^2 = 9$, $3^3 = 27$ combinations. Treatment df = $N - 1$.

2 Effects

Main effect	Average change from one factor's level change, over the other factors
Simple effect	Effect of one factor at a **particular level** of another
Interaction	Effect of one factor **depends on** the level of another

Parallel mean profiles = no interaction; non-parallel = possible interaction (needs a test).

3 2^2 factorial (r replications)

Main effect of A

$$\frac{(A)_{i=1} - (A)_{i=0}}{2r}$$

SS due to A

$$SS_A = \frac{[(A)_{i=1} - (A)_{i=0}]^2}{4r}$$

A × B interaction

$$\frac{(AB)_{i+j=0} - (AB)_{i+j=1}}{2r}$$

A, B and A×B each have **1 df**. In 2^3: divisors become $4r$ (effects) and $8r$ (SS); A, B, C, AB, AC, BC, ABC each have 1 df. **Yates' method** is the tabular algorithm for 2^n effects; Fisher's algebraic method is the alternative.

4 Degrees of freedom

Factor with a levels	$a - 1$
A × B	$(a-1)(b-1)$
3^2: A, B	2 each
3^2: A × B	4

A 3-level factor's 2 df split into **linear** $A_L = A_2 - A_0$ and **quadratic** $A_Q = A_2 + A_0 - 2A_1$.

5 Types of factorials

Symmetrical	All factors have the same number of levels (2^n, 3^n)
Asymmetrical	Unequal levels, e.g. 2×3
Complete (full)	All combinations included
Fractional	Only a selected fraction of combinations

6 Critical differences

Interaction means (2^2; ABC in 2^3)

$$t\sqrt{\frac{2\,MSE}{r}}$$

Two-factor means in 2^3

$$t\sqrt{\frac{MSE}{r}}$$

Levels of A in 3^2

$$t\sqrt{\frac{2\,MSE}{3r}}$$

Levels of B in 2×3

$$t\sqrt{\frac{2\,MSE}{2r}}$$

$t = t_{\alpha/2,\,edf}$. If the interaction is significant, compare combination means, not main-effect means.

7 Confounding

Deliberately making an interaction **inseparable from block effects** so that blocks can be smaller.

Complete
The **same** interaction is confounded in every replication.

Partial
Different interactions confounded in different replications.

GAIN smaller, more homogeneous blocks and better precision · **COST** lost information on the confounded effect. Higher-order interactions are usually the ones confounded.

! Common mistakes

- × Interpreting main effects when the interaction is significant.
- × Mixing up $2r$ (effect) and $4r$ (SS) divisors in 2^2.
- × Confounding a main effect.

? Fill in the blanks

1. Three nitrogen levels × two phosphorus levels give ____ treatment combinations. [6]
2. A 2^4 factorial has ____ treatment combinations. [16]
3. In a 3 × 4 factorial, the A × B interaction has ____ df. [6]
4. If the response to nitrogen changes with the irrigation level, the two factors show ____. [interaction]
5. In a 2^3 factorial, each main effect and interaction has ____ df. [1]
6. Confounding ABC in every replication is ____ confounding. [complete]

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